Traffic Light Control System – Corrected Engineering Specification

T-Junction Traffic Signal Controller | Verilog/RTL Implementation Reference

This corrected specification defines the road geometry, permitted movements, signal phases, safety assumptions, compatibility rules, and FSM requirements for an academic Verilog traffic-light controller.

1. Junction Geometry and Direction Convention

Signal A: WEST arm, vehicles travel West → East. Signal B: EAST arm, vehicles travel East → West. Signal C: SOUTH arm, vehicles travel South → North. There is no continuing North road, so C has no straight-through movement.

2. Defined Vehicle Movements

Signal	Vehicle direction	Movement	Designation
A	West → East	Straight toward East/B	A-G1
A	West → East	Right turn toward South/C	A-G2-Right
A	West → East	U-turn back toward West	A-G2-U
B	East → West	Straight toward West/A	B-G1
B	East → West	Left turn toward South/C	B-Free-Left
B	East → West	U-turn back toward East	B-G2-U
C	South → North	Right turn toward East/B	C-Green-Right
C	South → North	Left turn toward West/A	C-Free-Left

3. Engineering Safety Assumptions

• Green-2 phases contain turning and/or U-turn movements and are exclusive.

• Only the explicitly approved phase combinations in the compatibility table are allowed.

• All unlisted or ambiguous combinations are treated as unsafe and must not be generated by the FSM.

• Permissive/free-left movements are modeled as yield-controlled movements, not unconditional right-of-way movements.

• For this academic controller, free-left movements may be represented as active during the relevant signal states only if they are included in the approved phase definition.

• Every change between vehicle phases shall include an all-red clearance interval; amber/yellow may be inserted before all-red.

• This specification is for a modeled T-junction and is not a substitute for a site-specific road-safety audit or statutory traffic approval.

4. Corrected Signal Phases

Phase	A	B	C	Permitted movements
P0 – ALL_RED	R	R	R	Clearance; no protected movement
P1 – A_G1	G1	R	R	A straight East
P2 – B_G1	R	G1	R	B straight West
P3 – AB_G1	G1	G1	R	A straight East and B straight West in parallel
P4 – C_GREEN	R	R	G	C right to East; C left is permissive/yield-controlled if enabled
P5 – A_G2	G2	R	R	A straight + right to C + U-turn
P6 – B_G2	R	G2	R	B straight + U-turn; B left to C only under defined yield rule
P7 – AMBER/CLEAR	Y or transition	Y or transition	Y or transition	Transition state followed by P0

5. Corrected Compatibility Matrix

A state	B state	C state	Allowed?	Reason
R	R	R	YES	All-red clearance
G1	R	R	YES	A straight only
R	G1	R	YES	B straight only
G1	G1	R	YES	Opposing straight movements; parallel phase
R	R	G	YES	C protected phase
G2	R	R	YES	A exclusive turning/U-turn phase
R	G2	R	YES	B exclusive turning/U-turn phase
G1	R	G	NO	Not approved; possible interaction/merge
R	G1	G	NO	Not approved; possible interaction/merge
G1	G1	G	NO	Three-way activation prohibited
G2	G1	R	NO	Green-2 must be exclusive
G1	G2	R	NO	Green-2 must be exclusive
G2	G2	R	NO	Two Green-2 phases prohibited
G2	R	G	NO	Green-2 must be exclusive
R	G2	G	NO	Green-2 must be exclusive
G2	G2	G	NO	Unsafe and prohibited

6. FSM State Sequence

P0 → P1 → P0 → P2 → P0 → P3 → P0 → P4 → P0 → P5 → P0 → P6 → P0

The order may be changed according to demand inputs, but every transition must pass through clearance. The FSM must never generate an unapproved combination.

7. RTL Output Encoding

• R = red lamp active

• Y = amber/yellow lamp active

• G1 = protected straight-green indication

• G2 = protected turning/U-turn-green indication

• G = green indication for Signal C

• At most one of G1/G2/G may be active per signal at a time.

• Whenever any signal is in G2, all other signals must be RED.

• During P0, all vehicle signals must be RED.

• Illegal/default FSM states must force P0/all-red outputs.

8. Verification Requirements

• Check that no prohibited matrix row is generated.

• Check amber/all-red clearance between phases.

• Check reset and illegal-state recovery go to P0.

• Check A-G1 and B-G1 together only in P3.

• Check C_GREEN never overlaps A_G1, B_G1, or Green-2.

• Check Green-2 phases are exclusive.

• Inspect truth tables and GTKWave waveforms.

9. Final Design Rule

Safety rule: Any phase combination not explicitly listed as allowed shall be treated as illegal and shall force an all-red clearance state.

Revision: v5 – Corrected conservative engineering specification